VOIDSCOPE · RISK REGISTER AND HAZARD ANALYSIS

VoidScope risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade): project risks, preliminary hazard analysis and first-pass design FMEA for option A

VoidScope risk register and hazard analysis

The top project risks are a missing medical lead to review the water and air lines (R2) and a drip margin of only 12 per cent (R4); the top design FMEA items are F8 bite valve (RPN 60), F5 bending links and pins (RPN 48), F7 capillary and clamp (RPN 40). Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 means the fault is caught before use, 5 means it cannot be seen.

> **Safety:** VoidScope is rescue equipment that may supply water and air to a trapped person and carries a lithium battery. Nothing here shows that it is safe to use. It is an open engineering reference, not certified rescue equipment and not a medical device.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
R1	Camera, blower or battery parts differ from the modelled sizes and ratings	Bought parts are single-source and not yet quoted	Redesign of the tip bore, plate layout or lid stack; cost rises	3	3	Design lead	Confirm each item in VDS-DEC-001 'To confirm' before buying; keep the camera bore a parameter
R2	No co-design partner or medical lead reviews the water and air lines	Neither named candidate has been contacted	Water line cannot move past TRL 3 safely	3	5	Project lead	Approach the first candidate (VDS-PRB-001); no drill with a person at the tip before review
R3	Tip heat limit missed with the real camera	Camera power at the dimmed setting is not known (STM S14)	Tip too hot to touch at 45 C ambient	3	3	Design lead	Choose a camera with a dimmer and a datasheet power; thermocouple test at TRL 4 (PR-014)
R4	Drip ceiling margin eroded by a different capillary or bottle height	Margin to 5 mL/min is 12 per cent at 2 m and 40 C with an oversize bore (STM S7)	Water flows faster than intended	2	5	Design lead	Measure flow at TRL 4; keep the 1 m rule on the drill card
R5	Open design is mistaken for certified equipment	Public hardware files on a rescue topic	Misuse by an untrained team	3	5	Project lead	Status statement in every document and the README (PR-019); release gate before any public step
R6	Patent in articulating camera tips or combined probes	The patent screen is preliminary (VDS-PRC-001)	Release delayed or design changed	2	3	Project lead	Focused CPC search before public release; not legal advice
R7	Printed PA12 links fail in drops or bend cycles	New weak point of the tip (STM S6)	Tip stuck or lost in a void	3	3	Design lead	Drop and cycle tests at TRL 4; spare sheath and links in the kit
R8	Outside lathe work unavailable locally	Nose, collar, tail cap and tip housing need a lathe	Build delayed or a printed fallback needed	2	2	Builder	Printed or milled fallbacks noted in VDS-DEC-001 value engineering
R9	Cold or heat outside the assumed battery range	Capacity at -10 C taken as 70 per cent (STM S15, S17)	Run time under 4 h	2	3	Electrical lead	Run-down tests at TRL 4; carry the second pack

Preliminary hazard analysis

Table 2. Hazards (stops S1 to S9 are in the build plan, section 6).

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Aspiration of water	Survivor drowsy, injured or lying badly; flow too fast; wrong bottle height	Choking, chest infection, death	5	2	Medical lead directs every use; conscious person who can drink only; gravity drip only; capillary ceiling 5 mL/min sized for 2 m and 40 C; bite valve opens only when bitten; never connect a pump or pressure source (build plan S5, S7, drill card 5a)	Medium; depends on the drill and the medical lead; to review at TRL 4
H2	Over-pressure or dust in the void from the air line	Blocked outlets; blower too strong; dusty rubble	Barotrauma or choking on stirred dust	4	2	Blower chosen by its shut-off pressure, 0.9 kPa, run from the regulated converter with a speed cap (fix F5); filter on the intake; flow meter; start at the lowest setting and watch the picture (S4)	Low to medium; shut-off pressure to confirm from the datasheet and measure at a blocked tip
H3	Hot tip touches a survivor	Camera and LEDs dissipate heat in a sealed housing	Contact burn	3	2	Black anodised housing; LEDs dimmed by default; 44 C at 35 C ambient and 54 C at 45 C (STM S14, fixes F1 and F4); above 35 C the tip is kept off the survivor and steered aside when offering water	Medium; at the limit and to measure
H4	Lithium battery fire or venting	Short circuit, damage, charging below 0 C, wrong charger	Fire, burns, smoke in a rescue area	4	2	LiFePO4 chemistry; packs with built-in protection; fuse; matching charger; charge away from the kit on a non-combustible surface, never below 0 C (S2, S3, S8)	Low
H5	Collapse or rubble shift during use	Probe used as a lever; operator on unstable debris	Injury to operator or survivor	5	2	Operator stays outside the void; probe never a lever; the team's structural safety procedures (S6)	Medium; outside the design's control
H6	Sharp edges and rebar	Cut tube ends, rebar, broken glass	Cuts to the operator; probe or sheath torn	2	3	Edges broken; gloves; the sheath is a spare part	Low
H7	Steered tip bent when withdrawn	Wires are not self-locking; tip	Tip stuck, wire or link damage, probe stuck	3	3	Lock straight before in or out (S9); the hole edge straightens a bent tip; wire factor 3	Low to medium

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
		left bent in the hole					
H8	Lathe, tube cutting and printing during the build	Rotating work, chips, hot nozzle	Injury to the builder	3	2	Build plan stop S1; guards and eye protection	Low
H9	Contaminated water path	Reused tube, valve or bottle	Infection of a weakened survivor	4	2	Single-use water set; cleaning kit and procedure; replace in under 5 minutes (PR-011)	Low to medium
H10	Cable or hose snagged on rubble	Lines run outside the void	Probe pulled, camera lost	2	3	Lines inside the rod and guide tube; hose and cable in an umbilical sleeve	Low

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Rod sections	Carry the push and bending loads	Buckles or bends	Probe unusable; tip lost in the void	Push above 940 N; a stuck probe pulled hard	4	2	3	24	Handbook limits; load test at TRL 4; never use as a lever
F2	Snap button and coupler	Lock the sections	Button fails to engage or shears	Sections separate in the void	Wrong wall thickness; weak button	5	2	3	30	Pins shear 6 MPa against 139 MPa; confirm button strength at 300 N (VDS-DEC-001 item 4)
F3	Camera head	See in dark and dust	Sealing fails or the window fogs	No picture	IP68 is a supplier claim	4	3	3	36	Immersion and drop tests (PR-004)
F4	Tip housing	Hold the camera and shed its heat	Overheats at 45 C ambient	Burn risk; camera shuts down	3 W in a sealed 33 mm housing	4	3	3	36	Black anodise and dimmed LEDs (fixes F1, F4); thermocouple test (PR-014)
F5	Bending links and pins	Bend the tip 90 degrees	Link cracks or a pin walks out	Tip stuck bent or lost	Printed PA12; 2 mm pins	4	3	4	48	Drop and cycle tests; grub screws and sheath retain pins (STM S6)
F6	Steering wire	Pull the tip round	Wire breaks at the thumb stop	Tip loose; cannot steer	Thumb force above 30 N; assumed breaking load wrong	3	2	3	18	Factor 3 on an assumed 250 N; confirm the rope (item 10)
F7	Capillary and clamp	Cap the water drip at 5 mL/ min	Bore oversize or partly clogged	Flow too fast or too low	Tolerance; debris in the water	5	2	4	40	Flow measured at 1 m and 20 C before each use; strainer; bore confirmed when bought (item 6)
F8	Bite valve	Pass water only when bitten	Leaks under head or sticks shut	Water dribbles on a survivor, or none arrives	Cracking pressure below 19.6 kPa	5	3	4	60	Check the cracking pressure and the bite in drills with the medical lead (S2, item 7)
F9	Blower	Move 20 L/ min at 0.9 kPa at most	Shut-off pressure above the datasheet	Over-pressure in the void	Wrong or substituted part	5	2	4	40	Select by shut-off pressure; measure at a blocked tip before use (S4)
F10	Intake filter	Give clean air	Clogged or bypassed	Dusty air or low flow	Heavy dust	3	3	2	18	

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
										The flow meter shows low flow; spare filter
F11	Battery pack	Power for a shift	Capacity lost in cold; cell fault	Run time under 4 h, or fire	Cold; damage; wrong charger	4	3	3	36	Second pack; protection circuit; charge rules (STM S15, S17)
F12	Camera cable	Carry picture and voice	Crushed or pulled at the tail cap	No picture	Snag on rubble	3	3	3	27	Grommets and strain relief at the tail cap; spare cable
F13	Monitor and lid	Show the picture	Glare or a cracked screen	Picture unreadable	Sun; a drop	3	2	2	12	Lid clearance 19 mm by model; screen shade
F14	Water guide tube	Carry the water tube inside the rod	Kinks or tangles with the cable	Water set cannot be changed	Bore too tight for the lines	3	2	3	18	PTFE guide; change procedure in the build plan

Items with severity 5 (F2, F7, F8, F9) get a TRL 4 test whatever their RPN. Options B and C are not analysed to FMEA depth; their main hazards are recorded in VDS-OPT-001.